Multimedia Tools and Applications (2025) 84:18829–18854
https://doi.org/10.1007/s11042-024-19818-0

Disaster assessment from social media using multimodal deep learning

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Received: 18 March 2024 / Revised: 30 May 2024 / Accepted: 3 July 2024 /
Published online: 11 July 2024
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Abstract
Real-time global event detection particularly catastrophic events has benefited significantly due to the ubiquitous adoption of social media platforms and advancements in image classification and natural language processing. Social media is a rich repository of multimedia content during disasters, encompassing reports on casualties, infrastructure damage, and information about missing individuals. While previous research has predominantly concentrated on textual or image analysis, the proposed study presents a multimodal middle fusion paradigm that includes Cross-modal attention and Self-attention to improve learning from both image and text modalities. Through rigorous experimentation, we validate the effectiveness of our proposed middle fusion paradigm in leveraging complementary information from both textual and visual sources.The proposed intermediate design outperforms current late and early fusion structures, achieving an accuracy of 91.53% and 91.07% in the informativeness and disaster type recognition categories, respectively. This study is among the few that examine all three tasks in the CrisisMMD dataset by combining textual and image analysis, demonstrating an approximate improvement of about 2% in prediction accuracy compared to similar studies on the same dataset.Additionally, ablation studies indicate that it outperforms the best-selected unimodal classifiers, with a 3-5% increase in prediction accuracies across various tasks. Thus, the method aims to bolster emergency response capabilities by offering more precise insights into evolving events.

Keywords Disaster assessment · Multimodal deep learning · Social media

1 Introduction

The process of preparing for, responding to, and learning from the repercussions of big failures is known as disaster management. It encompasses how individuals and organizations cope

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with the human, material, economic, or environmental implications of a disaster. Natural catastrophes and military warfare have consistently created peaks in death and morbidity, marking human life throughout history.

Social media networks get billions of photos and words every second documenting a wide range of events in our immediate environment. It is now feasible to recognize and categorize occurrences worldwide in real-time thanks to large-scale image recognition and textual understanding as foundational tools.

During emergencies, crucial information for both first responders and the public often comes through image-text pairs. However, just before disasters, users may upload fragmented or contradictory information as events unfold, making it challenging to identify important occurrences accurately. This leads to difficulties in automatically assessing the severity of disasters. While computer vision algorithms and natural language processing techniques can classify data effectively, incomplete data presents a significant obstacle [1]. Therefore, augmenting data is essential to ensure accurate classification and enable effective learning from the information available.

Figure 1 displays tweets from three recent significant disasters and related photographs. It is observed that relying solely on one modality often results in the loss of critical information. For instance, even if the tweet text in Fig. 1(d) mentions an earthquake in Southern Mexico with a 7.1 magnitude, the extent of the damage caused by this earthquake cannot be



RT @Cal_OES: PLS SHARE: We're capturing wildfire response, recovery info here: https://t.co/r89LKpjLPj https://t.co/HiA1oQF2Ax

(a) California Wildfires



5am #Maria update: a little weaker.Tropical Storm Watches posted for eastern NC. No direct hit expected @FOX5Atlanta https://t.co/oTSbTbqTey

(b) Hurricane Maria



RT @ajwamood: #ajwamood : Harvey the first major hurricane to strike the #US since 2005 , causes at least 1 fatality https://t.co/RvDkebyAAC

(c) Hurricane Harvey



RT @CNN: Photos show a magnitude-7.1 earthquake's destruction in central Mexico https://t.co/2MnQogvOvb https://t.co/YHxKdnalRx

(d) Mexico Earthquake

Fig. 1 Tweet text and image pairs from different disaster events with complementary information

deduced from the language. However, upon examining the picture accompanying this tweet, the immense damage caused by the earthquake becomes evident.

In situations like this learning from different modalities of data can significantly enhance analysis [2]. A comprehensive learning with a nuanced understanding of the situation can be achieved by incorporating diverse sources such as images and text. This approach enables the model to cross-reference information, filling gaps [3] and validating data across multiple modalities.

Multimodal deep learning is a novel machine learning paradigm that combines numerous data kinds such as pictures, text, audio, and numerical data with multiple intelligence processing methods to provide better results. Such deep learning models analyze the situation better by understanding the situation effectively through many modalities since some stimuli are only present in particular modalities. Multimodal social media data analysis has received little attention lately, despite the triumphs of multimodal learning in other fields. To support humanitarian organizations' planning, mitigation, response, and recovery activities while leveraging social media data for social good specifically necessitates time-critical analysis of the multimedia information produced during a crisis event.

The motivation for employing multimodal learning in disaster assessment arises from the limited exploration of automatically detecting crisis events using both visual and textual information. Despite the acknowledged importance of utilizing AI for social good and the significant impact of social media, there has been minimal research on this specific intersection. The prevalence of social media in crisis contexts and the extensive history of interdisciplinary research in humanitarian crisis efforts underscore the need for a more comprehensive approach. Previous methods concentrating on either images or text alone have prompted the exploration of multimodal learning as a more holistic and effective strategy for crisis event detection.

To address these issues, the proposed work assesses the severity of the situation based on the following labels: 1) (a) Informativeness: whether the social media post may be used to provide humanitarian relief in an emergency, (b) Disaster Type: identifying the type of disaster, 2) Humanitarian Category: categorizing the specific type of humanitarian information provided in the tweet, and 3) Severity: determining the severity of the emergency based on the damage represented in the image and text. The contributions of the work are as follows:

- A novel, multimodal middle fusion based framework is introduced for classifying multimodal data in the crisis domain. The approach involves extracting image features from the picture and generating word embeddings for the text for each image-text combination.
- A cross-modal attention technique is proposed to fuse information from both modalities, followed by the utilization of a self-attention mechanism to generate a set of features with internal correlations, which are then used to derive the classification results. This cross-modal attention approach avoids transferring negative knowledge between modalities, while the self-attention mechanism helps the model learn the correlations within the fused data.

The paper is arranged in the following manner. The related studies that form a literature base for this work are discussed in Section 2, Section 3 puts forth the proposed multimodal deep learning architecture, Section 4 tabulates the results obtained and compares it to previous works, Section 5 includes a discussion of the results and an assessment of the impact of the research, Section 6 includes the conclusion, followed by Section 6.1, where the study challenges, limitations and future work are discussed.

Table 1 Related works

Authors and Year	Dataset and No. of Samples	Types of Modalities	Study Methodology	Type of Fusion	Study Limitations / Future Scope	Performance Achieved
Abavisani et al. (2020) [4]	CrisisMMD	Images and Text	• Textual features were acquired using BERT, while image features were extracted using DenseNet • Created a cutting-edge cross-attention module that can exclude false or useless information and merge just the pertinent data from modalities • A multimodal variant of stochastic shared embeddings, or SSE, was added to solve the problem of sparse training data	Late Fusion	SSE lacks in providing more interpretability in discerning underlying relationships between different features or indicators of disasters.	Accuracy obtained was in the range of 89-91% for various tasks.
Sirbu et al. (2022) [5]	CrisisMMD and tweets collected using Twitter API	Text	Adopted a semi-supervised method to augment unlabelled data into the dataset and improve classification accuracy.	No Fusion	The inclusion of semi-supervised classifiers increases the method's training time frame.	F1 scores of 90.5% for the informativeness task and 88% for the humanitarian task were attained.
Zou et al. (2021) [6]	CrisisMMD	Text and Images	• Image characteristics were retrieved using the VGG-16 architecture, while FastNet model was used to learn textual characteristics. • In the described fusion technique, the concatenated eigenvectors were passed through three fully connected layers and a SoftMax layer, thereby using the fused representation as input to a classification model.	Early Fusion	• Using classifiers such as GANs can enhance catastrophe severity detection. • Inaccurate results were produced by unbalanced data, particularly when determining the degree of damage.	Accuracy attained is 87.6%, 91.25-92.6% for informativeness and humanitarian categories, respectively.

Table 1 continued

Authors and Year	Dataset and No. of Samples	Types of Modalities	Study Methodology	Type of Fusion	Study Limitations / Future Scope	Performance Achieved
Aamir et al. (2021) [7]	The dataset was acquired from PyImage Search readers, who utilized Google Images	Images	Two blocks of convolutional neural networks were applied to identify and detect the intensity of natural catastrophes.	No Fusion	Data augmentation methods can be used to enrich the data and handle class imbalance problems.	An average precision of 97% is achieved.
Zhang et al. (2022) [8]	Weibo crawled data on Henan torrential rain	Text and image	• Employed LDA for subject identification • Combination of BERT and VGG 16 was used for text and image analysis	No Fusion	Observed misclassification of weather warnings due to lack of coherent data in tweets.	Accuracy: 0.93
Ofli et al. (2020) [9]	CrisisMMD	Text and image	• CNN was used for textual analysis while VGG-16 was used for image analysis • The final categorization made use of shared representations from both modalities	Early Fusion	• The damage severity task was assessed only on images • The study lacked in giving good results for cases where each modality gave conflicting information	• The informativeness task yielded accuracy, precision, and F1 scores of approximately 84% • The humanitarian categorization job yielded accuracy, precision, and F1 scores of approximately 78%
Belcastro et al. (2021) [10]	Data collected from Twitter and Synthetic Posts	Text	• Neural networks were utilised to filter gathered tweets to identify pertinent ones • By approximating the coordinate, geocoding services were used to enhance the tweets with precise location tags • Finally spatial clustering was employed to group events based on key problems	No Fusion	Identifying the severity of the disaster was beyond the scope of this work.	F1 Score greater than 85% was attained.

Table 1 continued

Authors and Year	Dataset and No. of Samples	Types of Modalities	Study Methodology	Type of Fusion	Study Limitations / Future Scope	Performance Achieved
Ponce-López, and Spataru (2022) [11]	Social Media Disaster Tweets (SMDT), Multilingual Disaster Response Messages (MDRM), UnifiedCEHMET	Text	• Employed deep bidirectional Transformer neural networks for feature selection and classification approaches like SVM and Naïve Bayes for multiclass classification • For binary classification BoW, TFIDF and DistilBERT were used	No Fusion	Bidirectional Transformer neural networks suffered higher complexities while learning features.	• Compared to Bernoulli Naïve Bayes, which showed precision values of 0.59 and 0.76, and Multinomial Naïve Bayes, which showed precision scores of 0.79 and 0.91, Support vector machines showed superior precision with scores of 0.83 and 0.79 • For both MDRM and Unified-CEHMET, DISTIL BERT outperformed Bag-of-Words (BOW) and TFIDF approaches in terms of medical, humanitarian criteria and severity • Concerning MDRM medical and humanitarian criteria, the accuracy rates for DISTIL BERT were 80% and 86%, respectively, whereas the accuracy for the Unified CEHMET severity assessment was 92%
Asif et al. (2021) [12]	CrisisMMD and customized dataset using Google Open Images Dataset (OID)	Images	• Disaster-related images were assessed to identify the kind of disaster and pertinent visual indicators within the images using the VGG-16 and You Only Look Once (YOLO) algorithms • Decision tables and applied hierarchical processes (AHP) were used to connect the intermediate results, putting images of disasters into a systematic taxonomy of disasters	No Fusion	The accuracy of the analysis may be improved by including social media spatiotemporal information in the algorithms.	VGG-16 achieved 83.6% testing accuracy, whereas YOLOv4 correctly identified 96% of photos in the disaster taxonomy.

Table 1 continued

Authors and Year	Dataset and No. of Samples	Types of Modalities	Study Methodology	Type of Fusion	Study Limitations / Future Scope	Performance Achieved
Ochoa and Comes (2021) [13]	EM-DAT data enriched with demographic and geographic information	Text, satellite images, statistical data, indexes	VGG-16 was used for feature selection, with unsupervised Self-Organizing Maps (SOM) commissioned to serve as a numerical predictor for estimating possible catastrophes.	Decision level Late Fusion	Attributes selected do not fully capture the complexities of the disaster.	Prediction accuracy achieved is 62%
Yang and Cervone (2019) [14]	Image Net data set for training and image data in USGS Hazards Data Distribution System (HDDS) Explorer Website for testing	Images	Inception V3 is utilized for feature extraction, and an ensemble max voting classifier was tapped on to forecast the degree of damage.	No Fusion	More features are required to assess the extent of damage to exactly quantify the severity of damage.	85.6% accuracy and 89.09% F1 score.
Khalaf et al. (2020) [15]	Sensory data from the Environment Agency	Sensor data values	The efficacy of various machine learning classifiers was compared with Ensembles of LSTM	No Fusion	Optimization methods such as PSO and GA can be used to regulate the findings.	Long-Short Term Memory model and random forest ensemble learning fared better than individual models with 71.4%, 85.9%, and 81.13% sensitivity, specificity, and accuracy, respectively.
Jena et al. (2020) [16]	USGS and Statistics Indonesia	Text (Earthquake estimating factors)	• The clustering method was employed to detect earthquake-susceptible areas. • The likelihood of earthquakes was assessed via the CNN model. • Risk was estimated and Analytic Hierarchy Process was implemented to estimate liabilities.	No Fusion	This study is restricted to a few key variables; due to data inaccessibility, soil properties, liquefaction factors, fault features, and precursors were left out.	The study achieved 89.47% accuracy for estimating the likelihood of an earthquake, and a consistency ratio of 0.07 for estimating its vulnerability.

2 Related works

A survey of the related works proposing unimodal and multimodal approaches for disaster assessment is tabulated in Table 1.

3 Methodology

The proposed multimodal model architecture is illustrated in Fig. 2, and the model layers and parameters are tabulated in Table 5.

3.1 Data collection

The proposed methodology is trained and tested on the publicly-available CrisisMMD [17] dataset. This multimodal dataset consists of tweets and associated images collected during seven different natural disasters that took place in 2017: Hurricane Irma, Hurricane Harvey, Hurricane Maria, the Mexico earthquake, the California wildfires, the Iraq-Iran earthquakes, and the Sri Lanka floods. The description of the dataset with the feature labels is as follows:

1. General Category: It is sub-tasked in the following manner.
   (a) Informativeness (Informative and Non-informative)
   (b) Disaster Type (Earthquake; Flood; Hurricane; Non-informative; Wildfire)
2. Humanitarian Categories (Affected individuals; Infrastructure and utility damage; Not humanitarian; Other relevant information; Rescue, volunteering, or donation effort)
3. Damage Severity Assessment (Little or no damage; Mild damage; Severe damage)

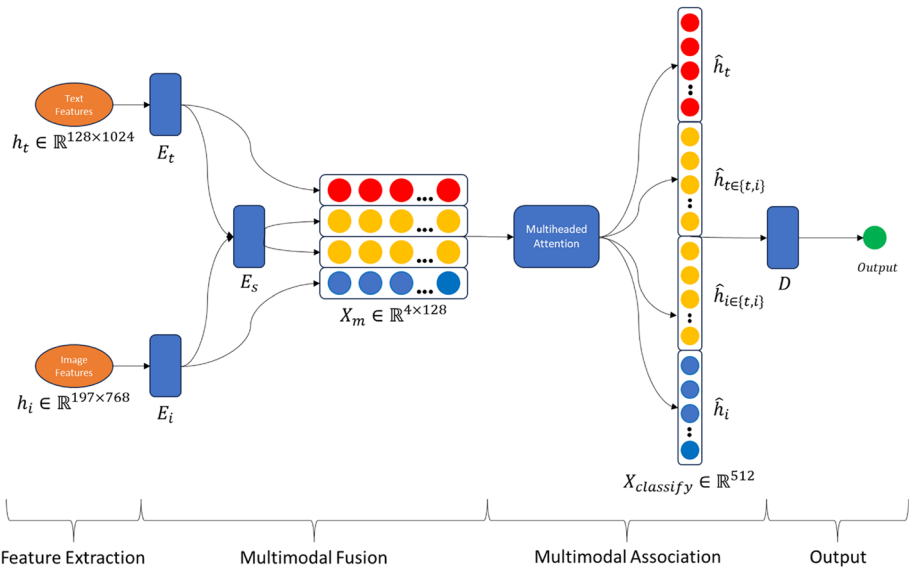


Fig. 2 Proposed multimodal model architecture

Table 2 Data distribution of the CrisisMMD dataset

Task	Label	Text	Image
Task 1(a)	Informative	11509	9374
	Non-Informative	4549	8708
Task 1(b)	Earthquake	1198	1372
	Flood	245	315
	Hurricane	7329	8307
	Non-informative	4549	8708
	Wildfire	1048	1127
Task 2	Affected individuals	472	562
	Infrastructure and utility damage	1210	3624
	Not humanitarian	4549	8708
	Other relevant information	5954	2529
	Rescue, volunteering, or donation effort	3293	2231
Task 3	Little or no damage	–	475
	Mild damage	–	839
	Severe damage	–	2212

The distribution of text and images in the CrisisMMD dataset are tabulated in Table 2. The dataset includes tweet text and image pairs with independently annotated labels for each task. However, only a subset of the dataset with the same label for both text and image pairs is used in the study. For the damage severity assessment task, which only includes labels for the images, the labels from the images are used for the corresponding tweets as well [4, 18, 19].

Table 3 puts forth the total training, test, and validation split for various tasks experimented on in the study. Detailed representation of no. of samples used in the study is presented in Table 4.

3.2 Data preprocessing

3.2.1 Text

Text from tweets frequently contains noisy data, such emoticons, symbols, and invisible letters. Therefore, using NLTK libraries, the textual input was preprocessed to remove stop words, URLs, hashtag signs, and non-ASCII characters. Whitespaces are used in place of the

Table 3 No. of samples and percentage of dataset used for all tasks

Task	Train	Dev	Test
Task 1(a)	9601 (75.55%)	1573 (12.38%)	1534 (12.07%)
Task 1(b)	9601 (75.55%)	1573 (12.38%)	1534 (12.07%)
Task 2	6126 (75.83%)	998 (12.35%)	955 (11.82%)
Task 3	2468 (70%)	529 (15%)	529 (15%)

Table 4 Data distribution of the data for each train, dev and test splits

Task	Label	Train		Dev		Test	
		Text	Image	Text	Image	Text	Image
Task 1(a)	Informative	5348	6345	1050	1056	1025	1030
	Non-Informative	2703	3256	511	517	499	504
Task 1(b)	Earthquake	766	919	152	152	130	130
	Flood	125	180	28	28	20	20
	Hurricane	3863	4587	757	763	725	730
	Non-informative	2703	3256	511	517	499	504
	Wildfire	594	659	113	113	150	150
Task 2	Affected individuals	62	71	9	9	8	9
	Infrastructure and utility damage	454	612	80	80	79	81
	Not humanitarian	2694	3252	519	521	500	504
	Other relevant information	1153	1279	238	239	233	235
	Rescue, volunteering, or donation effort	689	912	146	149	124	126
Task 3	Little or no damage	268	333	60	71	55	71
	Mild damage	480	587	113	126	100	126
	Severe damage	1380	1548	286	332	299	332

punctuation marks. Emoticons were swapped out for their associated meanings. The majority of misspelt words were corrected, and abbreviations were appropriately transcribed (Fig. 3).

3.2.2 Image

For image feature extraction from images, the input images are resized and cropped to 224x224, scaled to get pixel values in the range of 0 and 1 and each channel is normalized with respect to ImageNet data set.

3.3 Feature extraction

3.3.1 Text

The input text X_t is first tokenized [20, 21] and then encoded using the different variants of the BERT model, namely BERT [22], RoBERTa [23], and BERTweet [24]. This produces a sequence of text features h_t that capture the relevant semantic and contextual feature vectors of the input text. The ultimate selection for the conclusive multimodal architecture leans towards the RoBERTa variant, which exhibits the utmost accuracy in text classification. This choice is grounded in the understanding that the selected BERT variant has demonstrated superior performance, thereby ensuring the robust and precise integration of textual components within the broader multimodal framework. The working of the base BERT model is mathematically represented in (1).

$$h_t = RoBERTa(X_t) \tag{1}$$

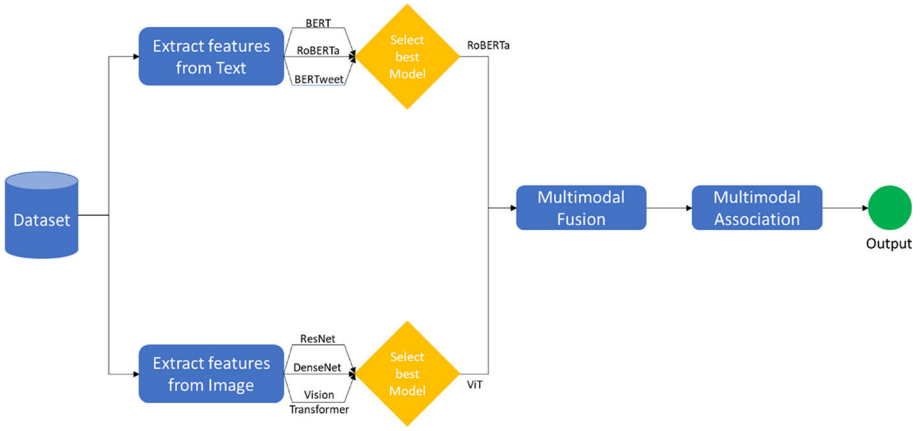


Fig. 3 Flowchart for unimodal model selection and multimodal model processing

where X_t is the input text and $h_t \in \mathbb{R}^{128\times1024}$ is the text feature map obtained from the RoBERTa model.

3.3.2 Image

The input image X_i is passed through an image encoder, which produces a set of image features h_i that capture the essential visual information in the input image. The rationale for selecting Visual Transformer (ViT) [25] over the CNN architectures [3] tested, ResNet [26] and DenseNet [27], stems from its higher performance, which is critical for effectively finding and selecting features required for disaster classification. This approach allows the strong integration of visual components required for identifying disaster-related patterns within the larger multimodal framework. The mathematical depiction of the operation of the basic ViT model is found in (2).

$$h_i = ViT(X_i) \tag{2}$$

where X_i is the input text and $h_i \in \mathbb{R}^{197\times768}$ is the image feature map obtained from the ViT model.

3.4 Multimodal fusion

The proposed work incorporates two primary forms of multimodal sequence data: Text (t) and Image (i). To fuse the features from both modalities, a cross-modal fusion attention mechanism is employed. Cross-modal attention is a fusion approach that leverages attention processes to blend features from many modalities. When complementary information is provided in many modalities, it is extremely useful. This fusion method has been prominent in recent cross-modal learning approaches [28–33].

In the data set that was employed, textual data gives context by explaining the location and severity of the damage, while images depict the extent of the damage visually. Through the use of cross-modal attention processes, these various forms of data may be effectively merged, enhancing our comprehension of the horrific event as a whole.

Furthermore, it facilitates semantic interpretation and raises the accuracy of assessments of the disaster scene by matching semantically related features across modalities. This helps identify the most relevant parts of the text given the image ($t\epsilon\{t, i\}$), and the most relevant parts of the image given the text ($i\epsilon\{t, i\}$).

In simpler terms, cross-modal attention captures connections between different types of information, like images and text. It helps understand how they relate to each other. For example, it can help identify the relevant regions in the images that correspond to the mentioned information in the text by attending to specific visual features.

To ensure that both sequences are of the same dimension, the output features are passed through feed-forward neural encoders, E_t and E_i, as represented in (3)-(5)

$$E_t(t) = tanh(Linear(tanh(Conv1d(t)))) \tag{3}$$

$$\bar{h}_t = E_t(h_t) \tag{4}$$

where encoder, E_t, is comprised of a 1d convolution layer and a linear layer with tanh activation function, and $\bar{h}_t \in \mathbb{R}^{128}$ is textual hidden vector.

$$E_i(i) = tanh(Linear(tanh(Conv1d(i)))) \tag{5}$$

$$\bar{h}_i = E_i(h_i) \tag{6}$$

where encoder, E_i, is comprised of a 1d convolution layer and a linear layer with tanh activation function, and $\bar{h}_i \in \mathbb{R}^{128}$ is image hidden vector.

A fully connected layer is used as an encoder to help identify relevant parts of each modality given the other (7).

$$E_s(s) = tanh(Linear(s)) \tag{7}$$

$$\bar{h}_{t\in\{t,i\}}, \bar{h}_{i\in\{t,i\}} = E_s(\{\bar{h}_t, \bar{h}_i\}) \tag{8}$$

where encoder, E_s, is comprised of a linear layer with tanh activation function, and $\bar{h}_{t\in\{t,i\}} \in \mathbb{R}^{128}$ and $\bar{h}_{i\in\{t,i\}} \in \mathbb{R}^{128}$ are the shared text and image hidden vectors.

Cross-modal attention uses the information interaction between the text and image modalities to produce a set of weighted features of both modalities that capture the important information in each modality. These representations are then stacked into a matrix, $X_m \in \mathbb{R}^{4\times128}$, as shown in (9).

$$X_m = [\bar{h}_t, \bar{h}_i, \bar{h}_{t\in\{t,i\}}, \bar{h}_{i\in\{t,i\}}] \tag{9}$$

3.5 Multimodal assocation

In the proposed work, the self-attentive mechanism of Transformer [34], also known as scaled dot product attention based on deflation, is adopted. By leveraging this attention mechanism, our model can effectively prioritize relevant information while downweighting less important elements, thus improving its ability to process and understand complex data patterns. This approach, based on the principles of deflation, enables the model to efficiently attend to key features within the input sequences, enhancing its overall performance in tasks such as disaster assessment where capturing intricate relationships between modalities is crucial for accurate analysis. It is calculated via (10).

$$Attention(Q, K, V) = SoftMax\{\frac{QK^t}{\sqrt{d_k}}\} \cdot V \tag{10}$$

Q, K, V are the query, key, and value matrices respectively.

The multimodal data with computed internal correlations, X_{Att}, is given by the (11).

$$X_{Att} = Attention(X_m) = [\hat{h}_t, \hat{h}_i, \hat{h}_{t\in\{t,i\}}, \hat{h}_{i\in\{t,i\}}] \tag{11}$$

where $X_{Att} \in \mathbb{R}^{4\times128}$ is the Transformer output, and $\hat{h}_t, \hat{h}_i, \hat{h}_{t\in\{t,i\}}, \hat{h}_{i\in\{t,i\}} \in \mathbb{R}^{128}$ are the computed internal correlation vectors.

By computing internal correlations, we aim to capture the interdependencies and relationships within the data, allowing our model to effectively understand the complex interactions between modalities. This multimodal representation serves as the foundation for our fusion strategy, enabling us to leverage the complementary information across modalities for tasks such as disaster assessment (Table 5).

A joint-vector, $X_{classify} \in \mathbb{R}^{512}$, is constructed by concatenating the attention output, X_{Att}, as given in (12).

$$X_{classify} = [\hat{h}_t \oplus \hat{h}_i \oplus \hat{h}_{t\in\{t,i\}} \oplus \hat{h}_{i\in\{t,i\}}] \tag{12}$$

3.6 Output

The multimodal fused feature data, $X_{classify}$, is computed by a decoder, D, to derive the classification results using (13).

$$D(x) = Linear(tanh(Linear(x))) \tag{13}$$

$$y = D(X_{classify}) \tag{14}$$

where decoder, D, is comprised of two linear layers with tanh activation function, $X_{classify} \in \mathbb{R}^{512}$ is the aggregated joint-vector and $y \in \mathbb{R}^{n_classes}$ is the classification result.

Table 5 Model layers and parameters

Section	Layers	Parameters
Input Layer	Text features (RoBERTa)	Height: 128, Width: 1024
	Image features (ViT)	Height: 197, Width: 768
Multimodal Fusion	E_t : 1D Convolution Layer	in_channels: 128, out_channels: 1, kernel_size: 1
	Linear Layer	in_channels: 1024, out_channels: 128
	E_i : 1D Convolution Layer	in_channels: 197, out_channels: 1, kernel_size: 1
	Linear Layer	in_channels: 768, out_channels: 128
	E_s : Linear Layer	in_channels: 128, out_channels: 128
Multimodal Association	Layer Normalization	normalized_shape: 128
	Transformer Layer	d_model: 128, nhead: 8
	Layer Normalization	normalized_shape: 128
Output Layer	Linear Layer	in_channels: 512, out_channels: 128
	Linear Layer	in_channels: 128, out_channels: n_classes

4 Results

4.1 Text modality

The text dataset was trained using three BERT models for a comparative study: BERT, RoBERTa, and BERTweet.

BERT [22], short for Bidirectional Encoder Representations from Transformers, has revolutionized natural language understanding by capturing contextual information from both preceding and subsequent words in a sentence.BERT can evaluate tweets containing information on disasters by taking into account the surrounding context, which helps it understand the nature, location, impact, and severity of the event. BERT has the ability to recognise the faint linguistic hints that are characteristic of content connected to disasters by pre-training on a wide corpus of text, including tweets. This allows BERT to recognise terms, expressions, and background data that denote crises or disasters.

BERT merely has to be fine-tuned for disaster assessment, usually by adding an output layer that is customized for the particular task at hand. Because it can rapidly and efficiently adjust its learned representations to the subtleties of disaster-related language without requiring significant task-specific architectural changes, BERT is an excellent choice for analysing tweets linked to catastrophes.

RoBERTa [23], building upon BERT's foundation, enhances pretraining techniques to better understand language context. RoBERTa has better optimizations which ensures that it is more effective at understanding the subtle language used in tweets. Some of the refinements in RoBERTa are removing the Next Sentence Prediction (NSP) task, introducing dynamic masking, and utilizing larger batch sizes which offers better contextual understanding in events such as disaster assessment.

BERTweet [24] is an adaptation of BERT fine-tuned and trained on tweets to learn tweet-specific features and linguistic nuances. In addition, BERTweet, unlike regular BERT or

Table 6 Model performance comparison for text modality

Task	Model	Accuracy	Precision	Recall	F1 Score
1(a). Informative vs. Not Informative	BERT	86.31%	0.86	0.86	0.87
	RoBERTa	88.53%	0.88	0.88	0.88
	BERTweet	87.81%	0.88	0.88	0.88
1(b). Disaster Type	BERT	85.33%	0.86	0.85	0.85
	RoBERTa	87.81%	0.88	0.88	0.88
	BERTweet	87.22%	0.88	0.87	0.87
2. Humanitarian Categories	BERT	82.72%	0.82	0.83	0.82
	RoBERTa	85.13%	0.85	0.85	0.85
	BERTweet	85.34%	0.86	0.85	0.85
3. Damage Severity Assessment	BERT	64.08%	0.64	0.57	0.76
	RoBERTa	65.03%	0.65	0.59	0.71
	BERTweet	62.76%	0.63	0.23	0.48

RoBERTa, includes tweet-specific pre-processing stages such as an alternate mechanism for handling URLs, hashtags, and mentions.

The models were trained with a batch size of 16 for 10 epochs with early stopping and a learning rate of 10^{-6}. Table 6 presents the accuracy, precision, recall, and F1 score of the models on the test data. From Table 7, we can conclude that RoBERTa performs much better than BERT and BERTweet at the informativeness and damage severity tasks, and BERTweet performs better at the humanitarian task.

4.2 Image modality

To perform a comparative study, the image dataset was trained using two CNN models, ResNet-50 and DenseNet-121, and Vision Transformer [35].

ResNet-50 [26], a variant of the Residual Neural Network (ResNet) architecture, is designed to tackle the challenge of training very deep networks. It acts as a feature extractor, identifying important visual features in disaster images. By traversing its layers, ResNet-50 extracts hierarchical features ranging from low-level details like edges to high-level concepts such as objects and scenes. This feature extraction is beneficial for tasks like classifying or detecting disaster-related elements. Pretraining ResNet-50 on large-scale image datasets like ImageNet allows it to learn generic visual representations. Fine-tuning on a smaller set of disaster images further customizes its learned features to better suit disaster scenes. ResNet-50 excels in object detection and localization tasks, enabling the identification and classification of damaged buildings, infrastructure, or vehicles within disaster imagery. Additionally, ResNet-50 aids in scene understanding by categorizing images into different types or severity levels of disasters. This categorization facilitates the development of efficient response strategies.

DenseNet-121 [27], a variant of the Densely Connected Convolutional Network (DenseNet) family, is also analysed in this study to test its efficacy in capturing vital disaster-related image features. DenseNet-121 employs dense connectivity, where each layer receives input not just from its immediate predecessor but from all preceding layers. This dense inter-layer connection fosters enhanced feature reuse and more efficient information flow throughout the network. Unlike ResNet-50, which utilizes skip connections for layer

Table 7 Model performance comparison for image modality

Task	Model	Accuracy	Precision	Recall	F1 Score
1(a). Informative vs. Not Informative	ResNet-50	82.20%	0.84	0.82	0.82
	DenseNet-121	80.38%	0.82	0.80	0.81
	Vision Transformer	87.48%	0.87	0.87	0.87
1(b). Disaster Type	ResNet-50	75.23%	0.76	0.75	0.74
	DenseNet-121	74.76%	0.75	0.75	0.76
	Vision Transformer	80.38%	0.81	0.80	0.80
2. Humanitarian Categories	ResNet-50	77.30%	0.78	0.78	0.78
	DenseNet-121	71.15%	0.77	0.76	0.55
	Vision Transformer	82.72%	0.82	0.83	0.82
3. Damage Severity Assessment	ResNet-50	62.57%	0.64	0.63	0.63
	DenseNet-121	63.14%	0.64	0.63	0.63
	Vision Transformer	65.41%	0.66	0.65	0.58

interactions, DenseNet-121's connectivity pattern promotes a more direct propagation of information, potentially mitigating challenges related to vanishing gradients and facilitating effective learning, especially in deeper architecture

In the domain of disaster assessment through image analysis, Vision Transformers (ViTs) [25] present a novel approach distinct from traditional convolutional neural network (CNN) architectures like ResNet-50 and DenseNet-121. ViTs leverage the transformer architecture, originally designed for natural language processing tasks, to directly process image data as sequences of tokens. This fundamentally differs from CNNs, which process images through a series of convolutional and pooling layers. One significant difference between ViTs and CNNs lies in their attention mechanisms. ViTs utilize self-attention mechanisms to capture global dependencies between different parts of the image, allowing them to consider long-range interactions and contextual relationships. In contrast, CNNs like ResNet-50 and DenseNet-121 primarily rely on local receptive fields and hierarchical feature extraction, which may limit their ability to capture global context efficiently.

The models were trained with a batch size of 10 for 10 epochs with early stopping. The learning rate was optimized to minimize loss and gradient. Table 7 presents the accuracy, precision, recall, and F1 score of the models on the test data. From Table 7, we can conclude that Vision Transformer performs much better than ResNet and DenseNet at extracting relevant information from the images for disaster assessment.

4.3 Multimodal fusion

The model was built using the architecture proposed in Section 3. The model was trained with a batch size of 16 for 10 epochs with early stopping. Cross Entropy Loss was chosen as the loss function along with Adam optimizer [36] with a learning rate of 10^{-5}. To perform a comparative study of the performance of the proposed model, the accuracy, precision, recall, and F1 score of the proposed model, score fusion, and other best-performing models are tabulated on the test data in Table 8.

Table 8 Model performance comparison for multimodal fusion

Task	Model	Accuracy	Precision	Recall	F1 Score
1(a). Informative vs. Not Informative	Abavisani et al. [4]	89.33%	–	–	0.89
	Zou et al. [6]	87.60%	0.88	0.88	0.88
	Sirbu et al. [5]	–	0.91	0.91	0.91
	Score Fusion	91.26%	0.91	0.91	0.91
	Proposed Approach	91.53%	0.92	0.92	0.92
1(b). Disaster Type	Score Fusion	90.35%	0.90	0.90	0.90
	Proposed Approach	91.07%	0.91	0.91	0.91
2. Humanitarian Categories	Abavisani et al. [4]	91.14%	–	–	0.92
	Sirbu et al. [5]	–	0.89	0.88	0.88
	Score Fusion	89.32%	0.89	0.89	0.89
	Proposed Approach	89.63%	0.90	0.90	0.90
3. Damage Severity Assessment	Abavisani et al. [4]	72.65%	–	–	0.70
	Zou et al. [6]	68.90%	0.66	0.67	0.67
	Score Fusion	65.78%	0.63	0.66	0.57
	Proposed Approach	70.32%	0.70	0.70	0.69

Table 9 Summary of results - task 1(a)

Metric	Text - RoBERTa	Image - ViT	Multimodal Fusion
Accuracy	0.8853	0.8748	0.9153
Precision	0.8845	0.8738	0.9153
Recall	0.8853	0.8748	0.9153
F1 Score	0.8832	0.8742	0.9153
Area Under the Receiver Operating Characteristic curve (AUC-ROC)	0.9432	0.9285	0.9679
Area Under the Precision-Recall curve (AUC-PR)	0.9269	0.8909	0.9495
Logarithmic Loss	0.3051	0.3243	0.2657
Matthews Correlation Coefficient (MCC)	0.7348	0.7137	0.8079
Specificity	0.8233	0.8314	0.8927
Cohen's Kappa	0.7315	0.7134	0.8079
Balanced Accuracy	0.8543	0.8531	0.9039
Youden's J statistic	0.7085	0.7062	0.8079
Positive Likelihood Ratio	13.80	9.458	13.80
Negative Likelihood Ratio	0.2499	0.2295	0.1376

In score fusion, the probability outputs of the text-only and image-only classifiers are combined to get the final predictions. The score fusion technique is being used here as a baseline for comparing the performance of the proposed multimodal model.

The proposed approach is the best-performing model for task1(a), Informativeness, achieving a higher accuracy (91.53%) than the previously best-performing models.

Table 10 Summary of results - task 1(b)

Metric	Text - RoBERTa	Image - ViT	Multimodal Fusion
Accuracy	0.8781	0.8038	0.9107
Precision	0.8799	0.8087	0.9104
Recall	0.8781	0.8038	0.9107
F1 Score	0.8758	0.7990	0.9098
Area Under the Receiver Operating Characteristic curve (AUC-ROC)	0.9761	0.9160	0.9814
Area Under the Precision-Recall curve (AUC-PR)	0.9790	0.9421	0.9850
Logarithmic Loss	0.3431	0.5967	0.2891
Matthews Correlation Coefficient (MCC)	0.8157	0.6917	0.8628
Specificity	0.9350	0.8730	0.9449
Cohen's Kappa	0.8135	0.6907	0.8621
Balanced Accuracy	0.9128	0.6319	0.9184
Youden's J statistic	0.8131	0.6768	0.8556

Table 11 Summary of results - task 2

Metric	Text - BERTweet	Image - ViT	Multimodal Fusion
Accuracy	0.8534	0.8272	0.8963
Precision	0.8586	0.8185	0.8973
Recall	0.8534	0.8272	0.8963
F1 Score	0.8543	0.8210	0.8963
Area Under the Receiver Operating Characteristic curve (AUC-ROC)	0.9563	0.9178	0.9734
Area Under the Precision-Recall curve (AUC-PR)	0.9551	0.9510	0.9768
Logarithmic Loss	0.4678	0.5621	0.3582
Matthews Correlation Coefficient (MCC)	0.7762	0.7226	0.8367
Specificity	0.9344	0.8746	0.9349
Cohen's Kappa	0.7749	0.7202	0.8363
Balanced Accuracy	0.7833	0.6292	0.7995
Youden's J statistic	0.7878	0.7018	0.8312

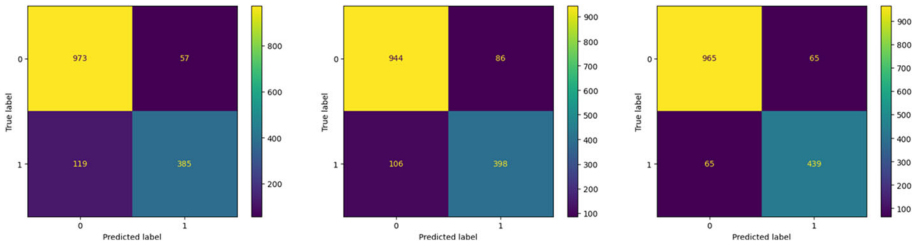
5 Discussion

5.1 Summary of results

This section compares the performance of the best-performing text and image models and the proposed multimodal fusion model for each task. The evaluation metrics include Accuracy, Precision, Recall, F1 score, Area Under the Receiver Operating Characteristic curve

Table 12 Summary of results - Task 3

Metric	Text - RoBERTa	Image - ViT	Multimodal Fusion
Accuracy	0.6503	0.6541	0.7032
Precision	0.5919	0.6610	0.6988
Recall	0.6503	0.6541	0.7032
F1 Score	0.5875	0.5754	0.6872
Area Under the Receiver Operating Characteristic curve (AUC-ROC)	0.6923	0.7814	0.7968
Area Under the Precision-Recall curve (AUC-PR)	0.7115	0.7958	0.8133
Logarithmic Loss	0.8457	0.7467	0.8266
Matthews Correlation Coefficient (MCC)	0.2317	0.2199	0.4097
Specificity	0.5215	0.5104	0.7127
Cohen's Kappa	0.1965	0.1765	0.4037
Balanced Accuracy	0.4438	0.3995	0.5450
Youden's J statistic	0.1718	0.1645	0.4159



(a) Text (b) Image (c) Multimodal Fusion

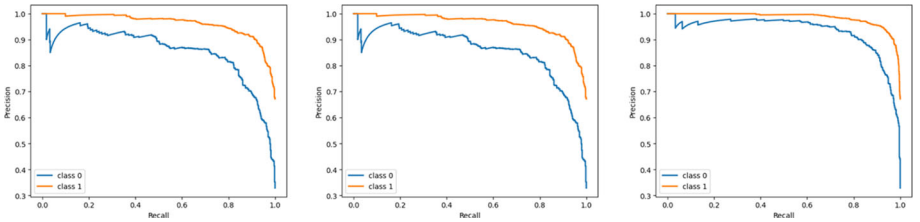
Fig. 4 Task 1(a): confusion matrix

(AUC-ROC), Area Under the Precision-Recall curve (AUC-PR), Logarithmic loss, Matthews Correlation Coefficient (MCC), Specificity, Cohen's Kappa, Balanced accuracy, Youden's J statistic, and Positive- Negative likelihood ratios.

Tables 9, 10, 11, and 12 show the performance of the text-only classifier, image-only classifier, and the proposed multimodal model on the various tasks. Figures 4, 6, 8, and 10 show the confusion matrices, while the Figs. 5, 7, 9, and 11 plot the Precision-Recall curve of the models on the various tasks (Figs. 6, 7, 8, 9, 10 and 11).

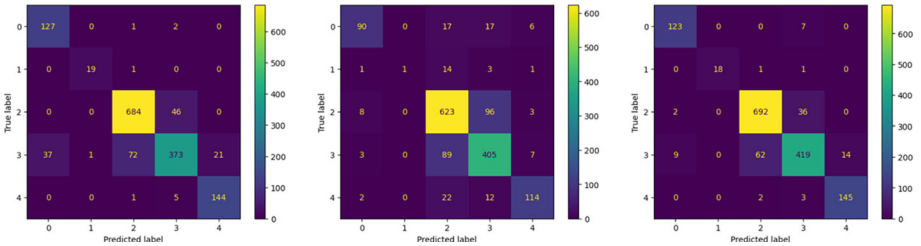
In general, the analysis of the text-only classifier reveals promising results across multiple evaluation metrics, with the performance of the image-only classifier being comparable. The text-only classifier performs better on informativeness and humanitarian tasks, suggesting that such information can be better inferred from language. While the image-only classifier performs better on the damage severity tasks, highlighting its advantage in identifying the severity of damage from images rather than textual data. However, the proposed multimodal model shows superior performance compared to both text-only and image-only classifiers.

Notably, the multimodal approach achieves higher accuracy, precision, recall, and F1 score, indicating its effectiveness in integrating textual and visual information for improved disaster assessment. AUC-ROC and AUC-PR metrics demonstrate improved discrimination between the class labels, reflecting the model's ability to leverage complementary features from both modalities. Logarithmic loss suggests reduced predictive uncertainty compared to individual modalities, while MCC signifies substantial agreement between predicted and actual outcomes. Specificity, Cohen's Kappa, balanced accuracy, and Youden's J statistic highlight the multimodal model's superior performance in correctly identifying non-disaster instances and distinguishing true negatives. Positive and negative likelihood ratios further underscore the model's capacity to differentiate between positive and negative class labels



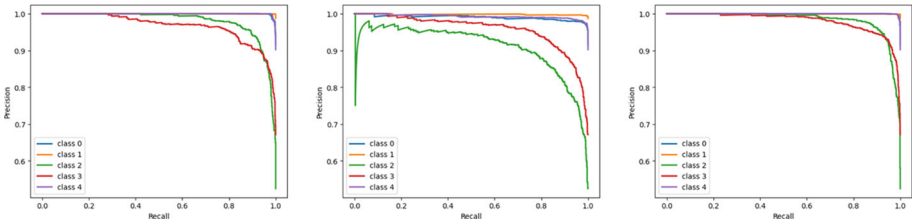
(a) Text (b) Image (c) Multimodal Fusion

Fig. 5 Task 1(a): precision-recall curve



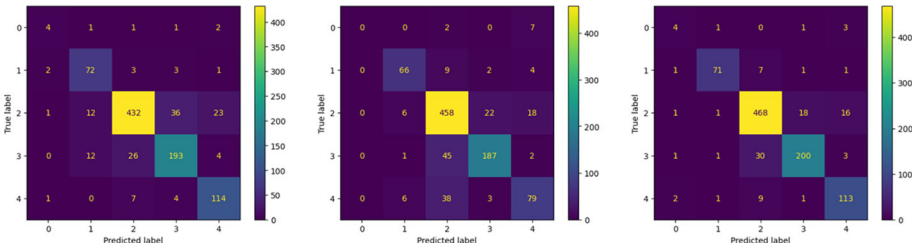
(a) Text (b) Image (c) Multimodal Fusion

Fig. 6 Task 1(b): confusion matrix



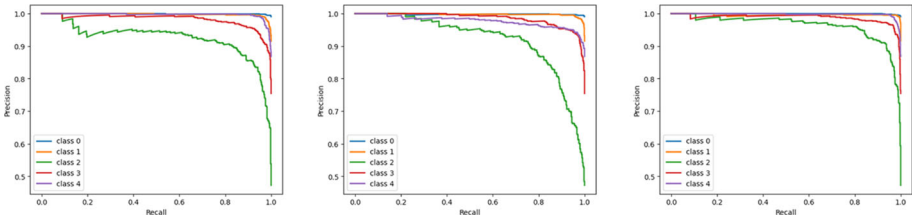
(a) Text (b) Image (c) Multimodal Fusion

Fig. 7 Task 1(b): precision-recall curve



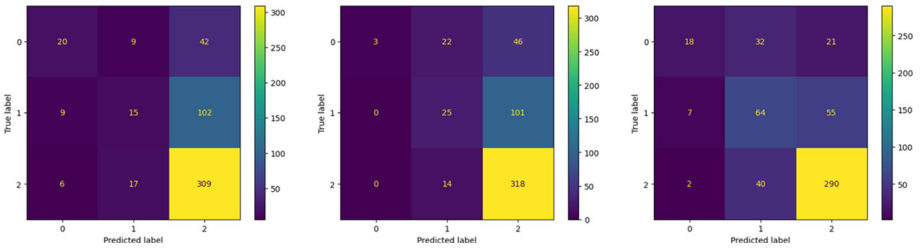
(a) Text (b) Image (c) Multimodal Fusion

Fig. 8 Task 2: confusion matrix



(a) Text (b) Image (c) Multimodal Fusion

Fig. 9 Task 2: precision-recall curve



(a) Text (b) Image (c) Multimodal Fusion

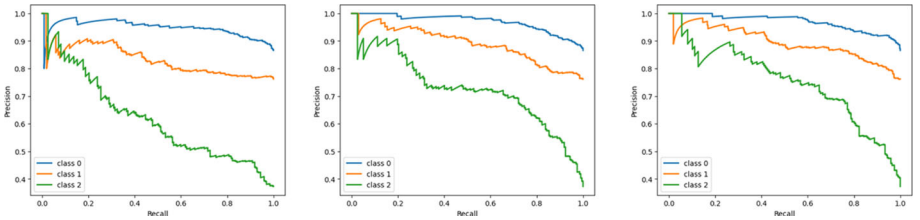
Fig. 10 Task 3: confusion matrix

with heightened certainty and reliability, affirming the efficacy of multimodal deep learning in disaster assessment.

5.2 Discussion

The proposed multimodal fusion technique is a novel approach that leverages both cross-modal and self-attention mechanisms. This innovation goes beyond previous approaches, which solely relied on cross-modal attention, to achieve superior information extraction at the intermediate fusion level. This section highlights the justification for employing the proposed key components in constructing the architecture. The justification is as follows:

1. Capturing Cross-Modal Dependencies: By using cross-modal attention, the model can efficiently incorporate input from both text and visual modalities. In disaster assessment circumstances, access to both visual (image) and contextual (textual) information is essential for a complete understanding of the event. By employing cross-modal attention, the model can capture the interdependencies between language and visual signals. This is achieved by allowing the model to focus on relevant portions of the image while also taking into consideration the textual descriptions or captions that correspond with those areas, and vice versa.
2. Self-Attentive Mechanism: Long-range dependencies can be captured within a single modality with the help of the self-attentive mechanism, also referred to as intra-modal attention. Through the application of self-attention in the combined text and image domains, the model can capture the inherent relationships and connections within each modality.



(a) Text (b) Image (c) Multimodal Fusion

Fig. 11 Task 3: precision-recall curve

3. Intermediary Feature Fusion: Cross-modal attention with self-attentive mechanism, combined with deflation and scaled dot product attention, serves as an intermediary feature fusion method. It allows the model to combine information from the image and text domains at a finer level, identifying connections and correlations between them. For instance, while an image may show debris and destruction, accompanying text might provide insights into the specific location or type of disaster. Such a kind of fusion enhances contextual understanding of the situation by mitigating the limitations owing to the ambiguous or incomplete information in any individual modality.

The results demonstrate that the proposed approach is particularly effective in capturing detailed dependencies within and between different modalities, enabling it to make more informed decisions, especially in disaster assessment tasks.

The proposed intermediate fusion provides a more contextually rich and nuanced representation than early and late fusion. Early fusion integrates modalities at the input level, potentially leading to information loss or simplifying of multimodal data. Late fusion, on the other hand, merges modalities at a higher level of representation, perhaps missing subtle interactions between them. Thus, in essence, intermediate fusion improves the interpretability and transparency of the model's decision-making process by providing a greater understanding of how different modalities affect the model's predictions. The proposed framework reflects the complicated interactions and dependencies across modalities required for tasks like damage detection and situational awareness. This thorough integration of multimodal data allows the model to make more accurate and informed judgments during calamities [37, 38].

5.3 Significance of the work

In this era of social media, enforcing the use of AI for disaster assessment within social media contributes paramountly to providing a more holistic understanding of the situation. For example, by analyzing both textual information and visual content, analysts can pinpoint areas with high levels of damage or where help is urgently needed. These approaches offer a thorough data analysis by leveraging information from multiple sources, thereby enhancing the model's performance even when dealing with incomplete data, which is often encountered in online social networks (OSNs).Most significantly, by cross-referencing information offered by multiple modalities, these multimodal fusion methodologies provide substantial information to disaster rescue teams, enabling timely assistance.

6 Conclusion

In this highly interconnected world, harnessing social networks to advance real-time crisis management operations is a breakthrough. The use of multimodal systems has shown to be quite useful in assessing disaster intensity. In the course of our research, we demonstrated that the proposed intermediate fusion approach outperforms early or late fusion methods when using multimodal learning models for tasks such as damage detection, resource allocation, and situational awareness during the crisis. By combining textual and visual data and facilitating the integration of information at a granular level, the model synthesizes a holistic knowledge of catastrophic occurrences through the integration of cross-modal attention, self-attentive mechanisms, and intermediary feature fusion.

The proposed research holds significant implications by offering a deeper comprehension of disaster scenarios, thereby substantially improving disaster response and recovery

endeavors. This enhanced understanding enables responders to make better-informed decisions, resulting in more efficient allocation of resources and coordinated efforts. Beyond disaster assessment, the proposed method holds promise for possible extension in diverse domains such as medical diagnosis, autonomous systems, and multimedia content analysis. In the field of medical diagnostics, utilizing data from many sources like scans, textual data, and patient records has tremendous potential for improving knowledge and aiding the early identification of diseases. Similarly, the proposed approach can be applied to improve decision-making in autonomous systems such as robots and surveillance. Furthermore, in multimedia content analysis, the proposed method can be extended to improve the relevance and accuracy of results for tasks such as content-based retrieval and cross-modal information retrieval.

6.1 Study challenges, limitations and future work

During the research, various challenges related to model training and dataset limitations were encountered. Transformer-based architectures used in the study, such as RoBERTa and Vision Transformer, showed a tendency to overfit. Additionally, the deep learning classifiers implemented were prone to catastrophic forgetting during knowledge transfer. To address these issues, a range of learning rates, regularization and other hyperparameters were explored to find the optimal configuration. Self-attention mechanisms were employed to mitigate the problems associated with incremental learning. Metrics such as validation loss were used to measure the performance of the classifiers [39]. The damage severity task heavily relies on visual information. Additionally, the data exhibits class imbalance, with fewer samples in the low and mid-severity classes compared to severe damage. Even in Task 2, the number of suitable message pairs containing both tweets and images was not found in equal proportions across all categories. These factors particularly led to poor performance of the methodology in Task 3 relative to other tasks. This limitation can be tackled in future research by implementing additional preprocessing techniques. These may include strategies like data augmentation, applying SMOTE for resampling, assigning appropriate class weights, generating balanced batches, and creating synthetic data using GANs.

There is a great degree of bias in research including data from online social networks because of differences in data representation, noise, and a dearth of ground truth labels. Furthermore, the abundance of misinformation in news-related data on social media presents considerable difficulty. An additional challenge to the analysis is the profusion of inconsistent data across different social network components. One significant shortcoming of the proposed study is its failure to handle cases in which both textual and visual information convey conflicting facts about a particular event. This study does not address the verification of misinformation conveyed through various modalities, if any.

Analyzing discriminative features or the correlation of characteristics in different modalities could address the aforementioned limitations. Techniques such as information relevance [40] can be explored to enhance model learning. To represent the complementary nature of visual and textual information in conveying information, studies could evaluate image-text relations and confirm these relations using appropriate metrics [41]. Additionally, AI-assisted labeling can be utilized to improve annotation quality, particularly when labels are inconsistent. Techniques such as t-Distributed Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) can be employed to validate inconsistencies in the data and preserve both local and global structures of the data.

Future research in this direction can include methods to improve the generalizability of the model. This might include adding additional interpretability to deep learning models or adopting explainable AI approaches [42] to give explanations or reasons for AI models' judgments. Additionally, future research could explore methods for verifying the accuracy of information and incorporating privacy metrics, such as ensuring the anonymity of users involved in accidents during disasters.

In upcoming research endeavors, more intricate multimodal fusion methods, such as - hierarchical structures, hybrid architectures and improved attention processes [43–45] that mainly improve image topic representations, can be investigated, to improve the accuracy of disaster assessments while incorporating textual and visual elements more effectively. Additionally, weakly supervised and semi-supervised learning strategies can be commissioned in the future to take advantage of the wealth of unlabeled social media data and label them, thereby reducing the dependence on manually labeled datasets. Furthermore, including additional dimensions like temporal and spatial values could offer valuable insights into improving disaster severity assessment.

Author Contributions Dr. Nisha P Shetty and Mrs. Jayashree Shetty conceived and designed the study. Mr. Yash Bijalwan and Mr. Pranav Chaudhari conducted the data analysis and coding. Dr. Balachandra Muniyal provided supervision for the research. The initial draft of the manuscript was penned by Dr. Nisha P Shetty, with input from all authors on earlier drafts. All authors reviewed and endorsed the final version of the manuscript.

Funding Open access funding provided by Manipal Academy of Higher Education, Manipal No funds, grants, or other support was received for conducting this study.

Data and Code Availability The data supporting the conclusions of this study can be accessed at https://crisisnlp.qcri.org/crisismmd. The code can be obtained from the corresponding authors upon a reasonable request.

Declarations

Competing Interests The authors have no competing interests to declare that are relevant to the content of this article.

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